Day-07 Calculating Backpropagation by Hand.

mathmatics Assignment:

problem 18 The Computational Graph (Forward pass).

Consider a mathmatic model representing a single neuron's before activation:

$$f(x, y, z) = (x+y).z$$

To evaluate this as a computational graph, we break it down into intermediate variable, let $q = x+y$, making the final function $f = q \cdot z$.

you are given the input values: $x=-2$, $y=5$ and $z=-4$.

Task A: Draw the Computation graph for this function.

TaskB: Execute the forward pass. Calsi the exact numerical values for the intermediate node q, and ~~info~~ the final output node f.

Solution:

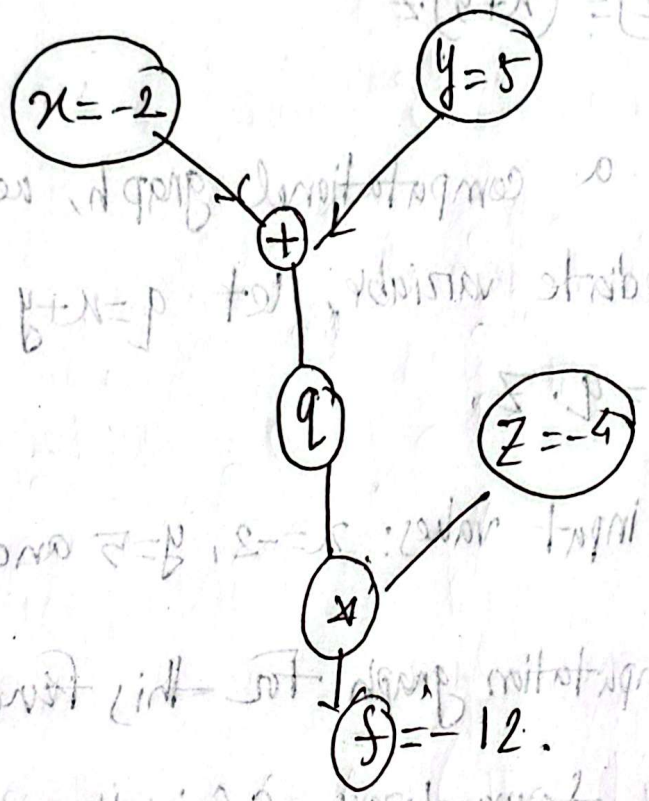
Task A/B: Draw the computation graph (Forward pass).

Given, $x = -2$, $y = 5$, $z = -4$

$$q = x + y = 3$$

$$f = q \cdot z = 3(-4) = -12$$

Result: $q = 3$, $f = -12$.



<u>problem 2</u>: The Backward pass (step by step Gradients)

we know, tiny change in our i/p (x, y, z) affects our final output f. we do this by passing the gradient backward from the output to the inputs.

Task A: (The output gradient): What is the derivative of the output w.r.t to itself, $\frac{\partial f}{\partial f}$?

Task B (Local gradient of Multiplication): look at the node $f = q \cdot z$. Calci the local Analytical gradients $\frac{\partial f}{\partial q}$ and $\frac{\partial f}{\partial z}$ and evaluate them using your forward pass numbers.

Task C (Local gradient of Addition): look at the node $q = x + y$. Calculate the local analytical gradients $\frac{\partial q}{\partial x}$ & $\frac{\partial q}{\partial y}$.

Task D (The chain Rule): Using the down stream gradient formula (Down stream = up stream x Local), Calci the final gradients with respect to the original input $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$

Solution:

Task A: $\frac{\partial f}{\partial f} = 1$

Task B: $\frac{\partial f}{\partial q} = -4$

$\frac{\partial f}{\partial z} = 3$

Task C: $\frac{\partial q}{\partial x} = 1$, $\frac{\partial q}{\partial y} = 1$

Task D:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} = -4$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial y} = -4$$

Problem 3: matrix calculus (The Linear layer)

Consider a Linear eqn: $Z = Wx + b$

where x is $n \times 1$ input

W is $m \times n$ matrix

b is $m \times 1$ bias vector

And Z is $m \times 1$ output vector.

Task A: Using the denominator layout, what is the gradient of Z w.r.to the input vector x. ($\nabla_x Z$)?

Task B: What is the gradient of w.r.to the weight matrix W, ($\nabla_W . W$)?

Solution: Task A:

$$\nabla_x Z = W^T.$$

To satisfy matrix multiplication dimensions moving backward, the $m \times n$ weight matrix must be transposed to $n \times m$.

Task B:

$$\nabla_W Z = X^T.$$

The gradient of weight is depend directly on the transpose input vector.

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